

Question 1

2 marks

101

Find the coterminal angle to $\frac{27\pi}{5}$ over the interval $[-360^\circ, 0^\circ)$.

Question 2

3 marks

102

Solve the following equation over the interval $0 \leq \theta < 2\pi$.

$$(\tan \theta - 3)(\tan \theta + 1) = 0$$

Question 3

2 marks

103

An earthquake in Vancouver had a magnitude of 6.3 on the Richter scale. An earthquake in Japan had a magnitude of 8.9 on the Richter scale.

How many times more intense was the Japan earthquake than the Vancouver earthquake?

You may use the formula below:

$$M = \log \left(\frac{A}{A_0} \right)$$

where M is the magnitude of the earthquake on the Richter scale

A is the intensity of the earthquake

A_0 is the intensity of a standard earthquake

Express your answer as a whole number.

Question 4

2 marks

104

Find and simplify the last term in the expansion of $(2y - 3x)^7$.

Question 5



3 marks

105

Given $\log_a 9 = 1.129$ and $\log_a 4 = 0.712$, find the value of $\log_a 12$.

Question 6

2 marks

106

How many different ways can 4 girls and 4 boys be arranged in a row if the girls and the boys must alternate?

Note: A calculator is not required for the remaining test questions.

Question 7

4 marks

107

Solve the following equation over the interval $[0, 2\pi]$.

$$2 \cos 2\theta - 1 = 0$$

Question 8

1 mark

108

Alex incorrectly explains to Rashid that the graph of $y = 2f(x) + 5$ means you first move the graph of $y = f(x)$ up 5 units and then multiply the y values by 2.

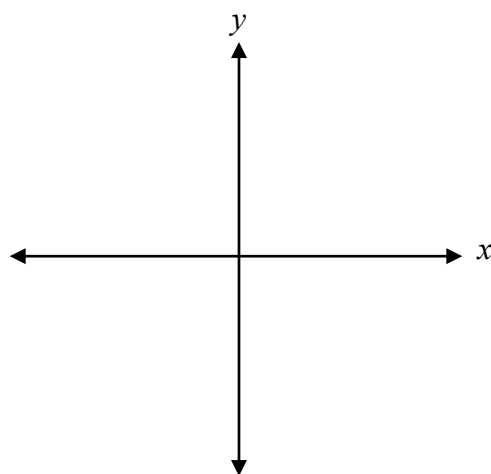
Explain to Rashid the correct way to transform the graph.

Question 9

1 mark

109

Sketch the angle of 5 radians in standard position.

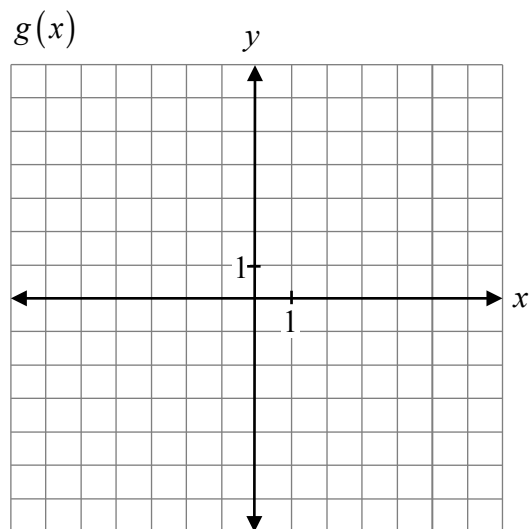
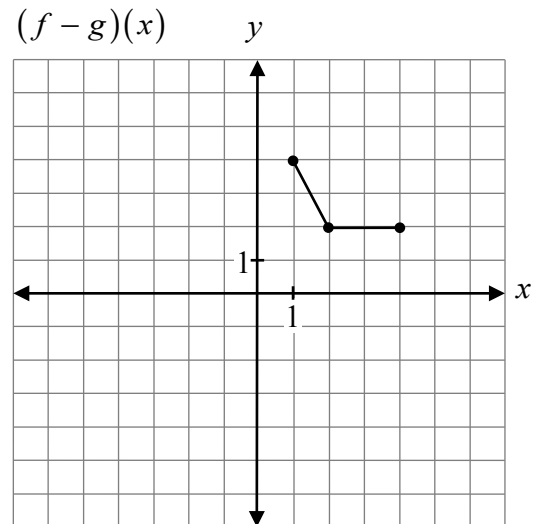
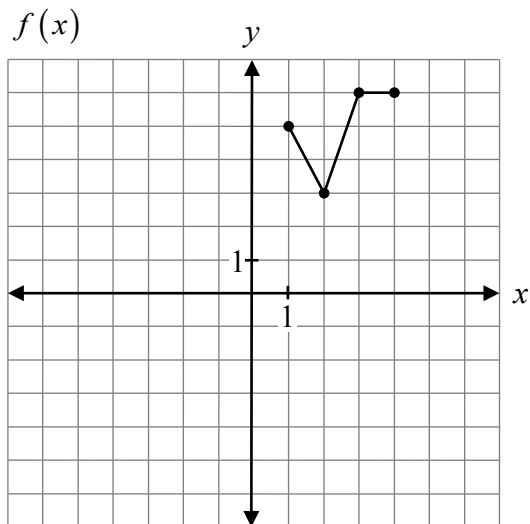


Question 10

2 marks

110

Given the graphs of $f(x)$ and $(f - g)(x)$, sketch the graph of $g(x)$.



Question 11

2 marks

111

A particular math class has a large number of students. From this class, you are to create a committee of 4 students that has at least 1 girl.

Without actually solving the problem, explain the strategy you would use to find the total number of ways to select this committee.

Question 12

a) 2 marks b) 2 marks

112
113

a) Prove the identity below for all permissible values of θ .

$$\frac{1 + 2 \cos^2 \theta}{\cos^2 \theta} = \tan^2 \theta + 3$$

Left-Hand Side	Right-Hand Side

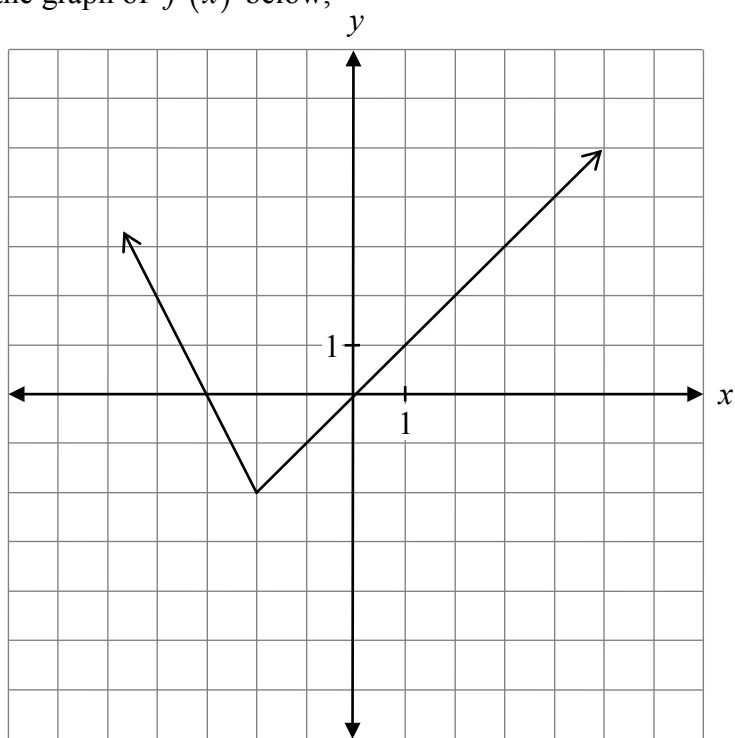
b) Determine all the non-permissible values for θ .

Question 13

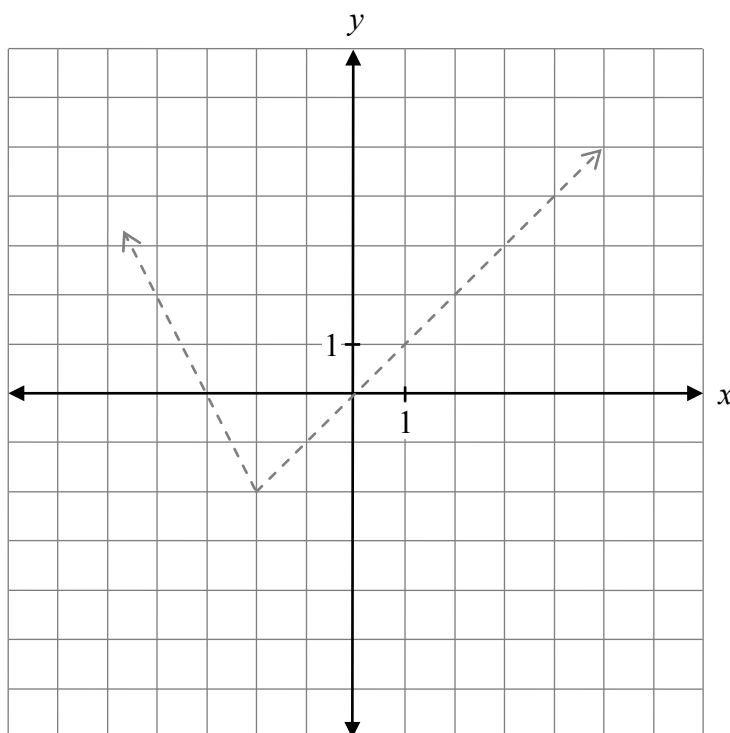
2 marks

114

Given the graph of $f(x)$ below,



sketch the graph of $g(x) = f(x - 2) + 3$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 14

a) 3 marks b) 1 mark

115
116

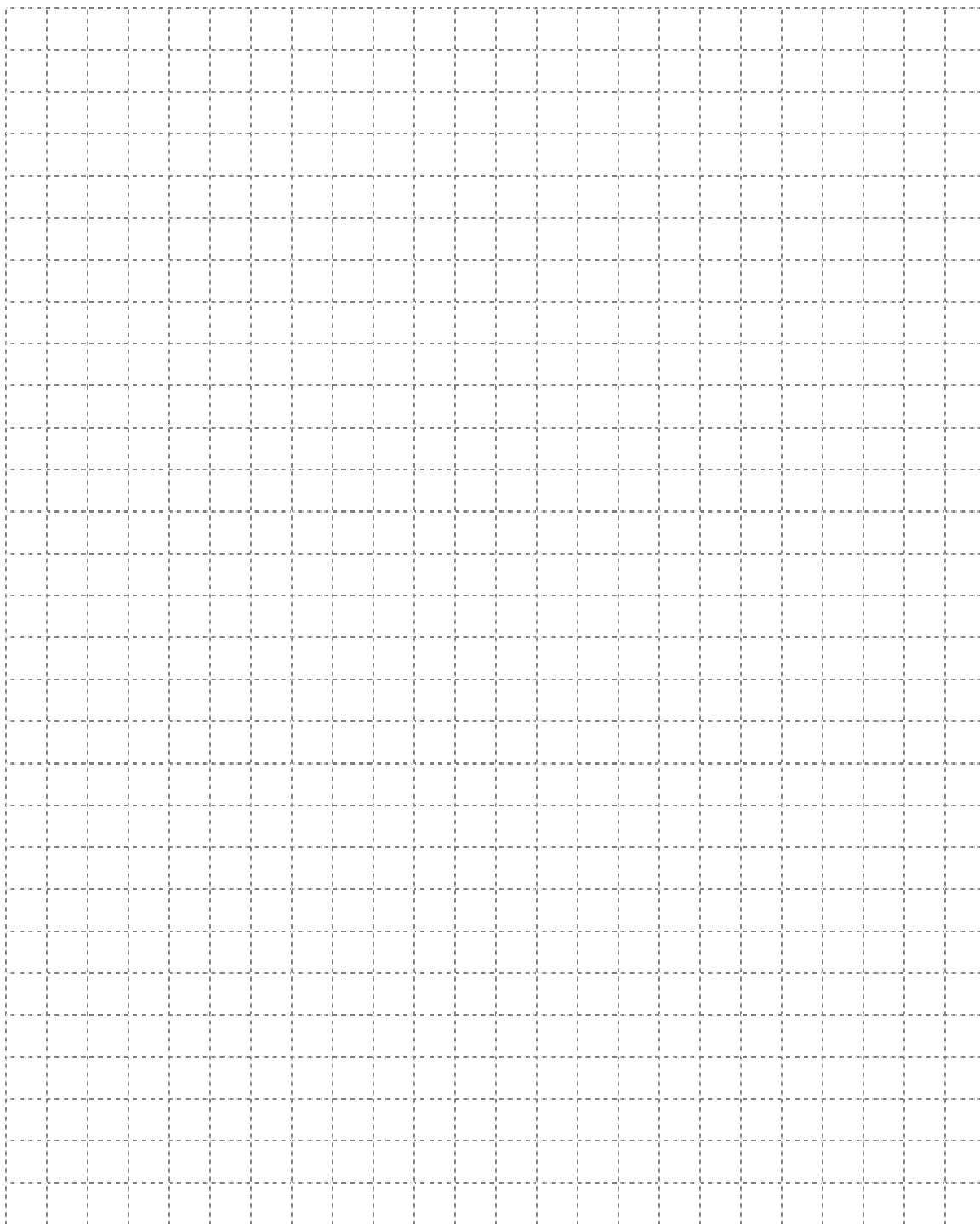
Given that $\sin \alpha = \frac{5}{13}$, where α is in Quadrant II, and $\cos \beta = \frac{2}{5}$, where β is in Quadrant IV, find the exact value of:

a) $\cos(\alpha + \beta)$

b) $\sin 2\alpha$

If $f(x) = x^3$ and $g(x) = 2x - 3$, what is the value of $\left(\frac{f}{g}\right)(-1)$?

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.